

Azure Services — RAG Lab Reference

Azure AI Foundry

Azure AI Foundry provides a workspace for developing generative AI applications and agents. It supports model selection, deployment, experimentation, evaluation, and application development.

Azure OpenAI / Azure AI Models

Azure-hosted AI models can provide capabilities such as text generation, summarization, reasoning, embeddings, and multimodal processing, depending on the model and service configuration.

In a RAG application, a text-generation model produces the final response while retrieval services provide supporting context.

Azure AI Search

Azure AI Search is a cloud search service used to index and retrieve information. It supports scenarios such as enterprise search, document retrieval, and RAG.

Important concepts include data sources, indexes, indexers, searchable fields, semantic ranking, and vector search.

Azure Storage

Azure Storage provides cloud storage services. Blob Storage is commonly used to store unstructured files such as PDFs, text files, images, and other documents.

In a document-based RAG workflow, a storage container can act as the location for source documents before they are processed and indexed.

Azure Virtual Machines

Azure Virtual Machines provide configurable compute resources for running operating systems and applications. A VM can be used as a development environment with tools such as Visual Studio Code, Python, Git, and Azure CLI.

VM networking, identity, security rules, and operating-system configuration should be considered when using a VM for development or training labs.

Azure Resource Groups

A resource group is a logical container for related Azure resources. Keeping lab resources in one resource group makes it easier to manage permissions, identify resources, and clean up the environment after training.

Azure Identity and Access

Azure role-based access control (RBAC) helps determine who can access Azure resources and what actions they can perform. Use the least-privilege approach: grant only the permissions required for the lab or application.

Example RAG Architecture

A simple Azure RAG solution can use Azure Blob Storage for source documents, Azure AI Search for indexing and retrieval, Azure AI Foundry for application and agent development, and a deployed language model for response generation.

Flow: User → AI application/agent → Azure AI Search → relevant document content → language model → grounded response.

Quick Reference

Azure AI Foundry: Build and manage generative AI applications and agents.

Azure AI Search: Index and retrieve enterprise or application knowledge.

Azure Blob Storage: Store source documents and unstructured data.

Azure Virtual Machine: Provide a configurable development or training environment.

Resource Group: Organize related Azure resources.